

IS THERE SIGNAL IN THE VASCULAR NETWORKS?

Spencer Venancio • Advisor: Dr. Anna Woodard
University of Wisconsin–Madison

Why vessels, and what we asked

Breast tumours are treated with chemotherapy *before* surgery. Patients who reach **pCR** — pathologic complete response, no residual invasive cancer at surgery — do markedly better, and knowing early who will get there would let clinicians change course while it still matters.

Tumours recruit their own blood supply, and **DCE-MRI** — dynamic contrast-enhanced MRI — images the breast repeatedly while a contrast agent washes through it. We ask whether the vessel network, not the tumour, carries the response signal.

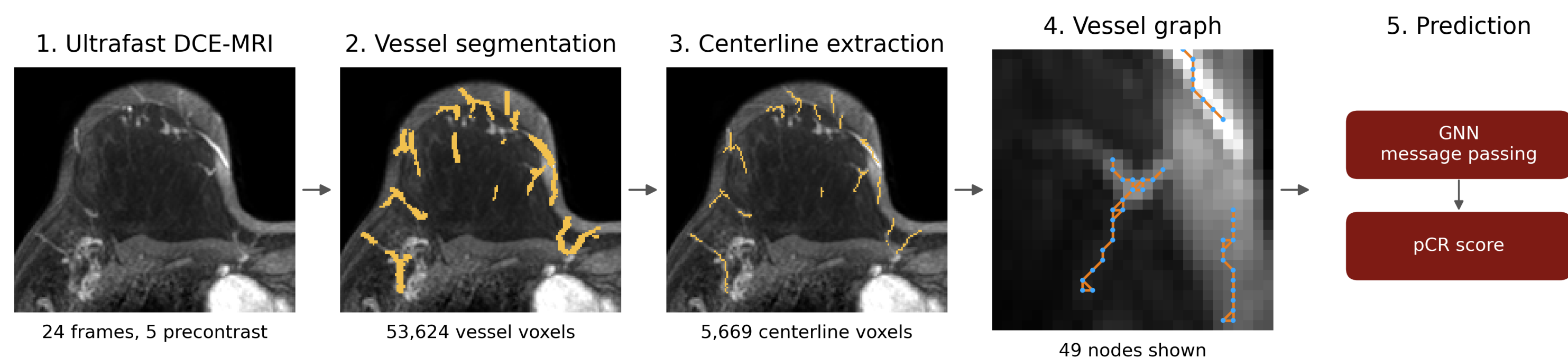
Is there signal in the vascular networks?

1 Is there signal in the blood vessels at all?

2 Does the graph add anything beyond that?

The two answers stay separate. Collapsing them into one “our model works” claim would hide which half is doing the work.

From scan to prediction



One real case: 24 frames over 119.5 s, 5 of them before contrast arrives.

A **centerline** is the one-voxel-wide skeleton running down the middle of each segmented vessel; every centerline voxel becomes a node of the patient’s vessel graph. Segmentation and centerline extraction are upstream lab pipeline — this project starts at the graph.

Fig. 1: Ultrafast DCE-MRI to vessel graph to prediction, on one real case. Panels 1–4 are measured data; panel 5 is schematic.

What the graph is

Nodes. Three representations, built from the same skeleton and compared on identical folds: one node per *centerline voxel*; one node per *vessel segment*; or nodes at *branch points and tips*.

voxel-as-node
node = every skeleton voxel

segment-as-node (line graph)
node = one whole vessel segment

junction-as-node, segment-as-edge
node = junction/endpoint, edge = segment

1 Signal in the vessels? YES

UChicago ultrafast DCE-MRI, $N = 179$, **vessels only** — no tumour mask, no clinical variables. Pooled out-of-fold AUC, case-resampling bootstrap CIs.

chance

2 Does the graph add? NOT YET

